

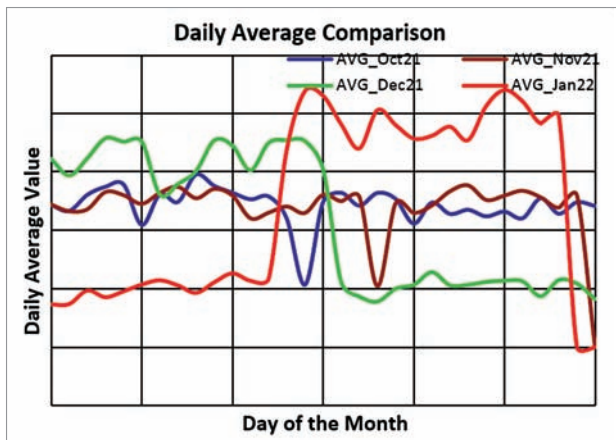


Fig. 6(a): The daily average values plotted for the four months under consideration

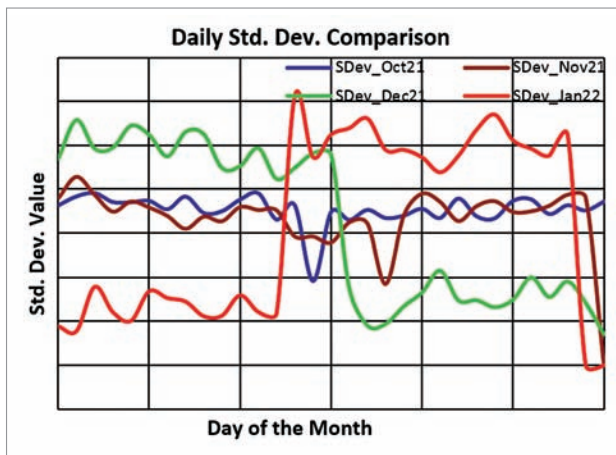


Fig. 6(b): The standard deviation values plotted for the four months

and normalised with respect to this monthly average. This retains the original nature of the curves, yet it becomes easier to compare data set using basic statistical analysis.

For the present study and data analysis, two months were primarily selected—January 2022 and December 2021. January is the month during which the actual volcanic eruption occurred in Tonga while the previous month of December 2021 was added to trace back for any prior clues. When significant indications were noticed in the data set for these two months, data from two previous months, October and November 2021, were added for comparison purposes. The data for the month of February 2022 could not be included due to extended

power interruptions during this month.

Understanding the observed data

Before we begin to analyse the data, let us first have a look at the daily signal variations. Fig. 4 shows the monthly averages for the selected four months of October 2021, November 2021, December 2021, and January 2022. On X-axis is the time of the day as recorded in Indian Standard Time (IST) clock. On Y-axis is the DC voltage representing the RF signal strength variation of a particular VLF channel or receiver.

It should be noted that each day there can be several variations, including earthquake precursors, solar bursts, etc in the signal plots.

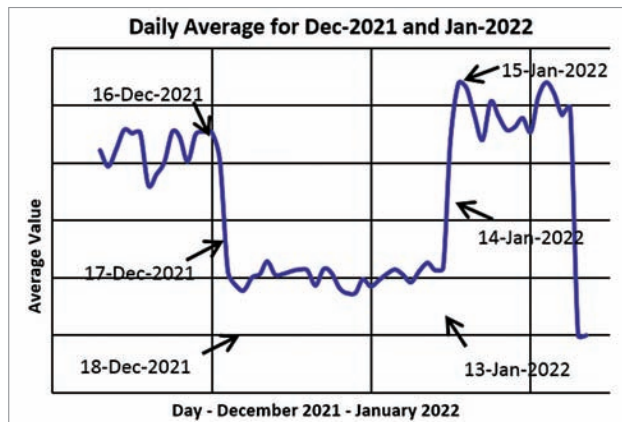


Fig. 6(c): The daily average values for December 2021 and January 2022 plotted on a continuous time scale reveal an exact and identical nature like an earthquake precursor

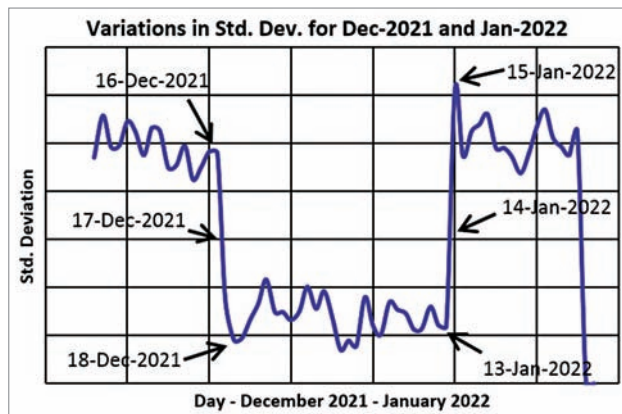


Fig. 6(d): The standard deviation values for December 2021 and January 2022 plotted here on a continuous time scale also reveal an exact and identical nature like an earthquake precursor

Therefore, each individual day will have different variations, but with the averaged data a smooth curve is obtained while the nature remains the same and the main features—such as the sun-rise-terminator (SRT), mid-day/noon peak, and sunset-terminator (SST)—can be identified easily.

Data analysis and interpretations

From the straight data plots for a day or even the monthly averages, it is not clear as to what changes have occurred in the signal variations over several days. However, if we plot all the days of a month, we can see some noticeable differences. Fig. 5(a) through Fig. 5(d) show all the day plots for the months of Octo-